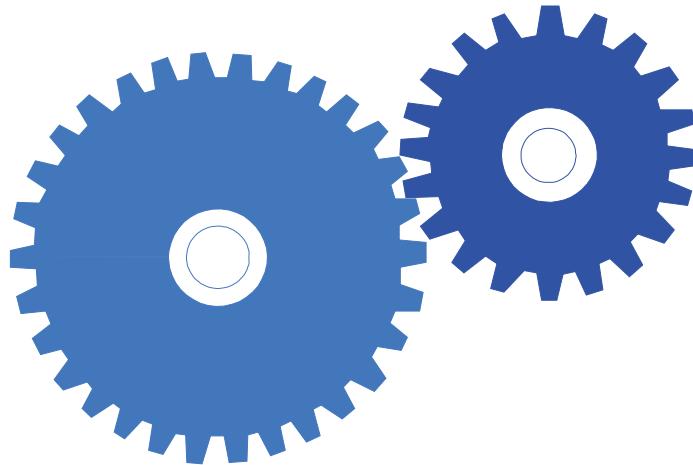


CHAPTER 4

OPERATION



CHAPTER 4

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SECTION 4-1

OPERATOR CONTROLS

 **CAUTION** Before operating, testing, or otherwise using any of the following components, read, understand and follow **Chapter 1, Safety**, and **Chapter 4, Operations, Sections 4-1, 4-3 and 4-4**.

This section covers the devices used by an operator to control **Peerless** mixers. See **Figures 4-1-1** and **4-1-2** for the standard control panel layout. **Chart 4-1-3, Operator Controls Summary**, lists a brief description of the components on a standard control panel.

SECTION 4-1

OPERATION CONTROLS SUMMARY CHART 4-1-1

COMPONENT NAME AND TYPE	FUNCTION	INTERLOCKS
DOUGH TEMP Digital thermometer	Indicates dough temperature.	None
EMERGENCY STOP Illuminated Push/Pull button	Stops all mixer functions in an emergency.	Must be pulled out for mixer to perform any function.
FLOUR GATE Selector Switch	Opens and closes flour gate.	Mixer will not start with gate open. Gate will not open if mixer is running. Bowl must be in full-up position.
HIGH SPEED Timer	Controls time mixer runs in high speed.	See Section 4-2 .
JOG Pushbutton	Jogs, or runs, agitator to discharge dough.	Must push SAFETY button at same time. End doors, flour gate and optional safety ingredient door must be closed.
JOG SELECTOR 2-position selector switch	Selects agitator speed to jog in.	None
LOWER Pushbutton	Lowers bowl.	Must push SAFETY button at same time. End doors, flour gate and optional safety ingredient door must be closed.
LOW SPEED Timer	Controls time mixer runs in low speed.	See Section 4-2 .
MIXER STOP Pushbutton	Stops mixer run cycle.	None

SECTION 4-1

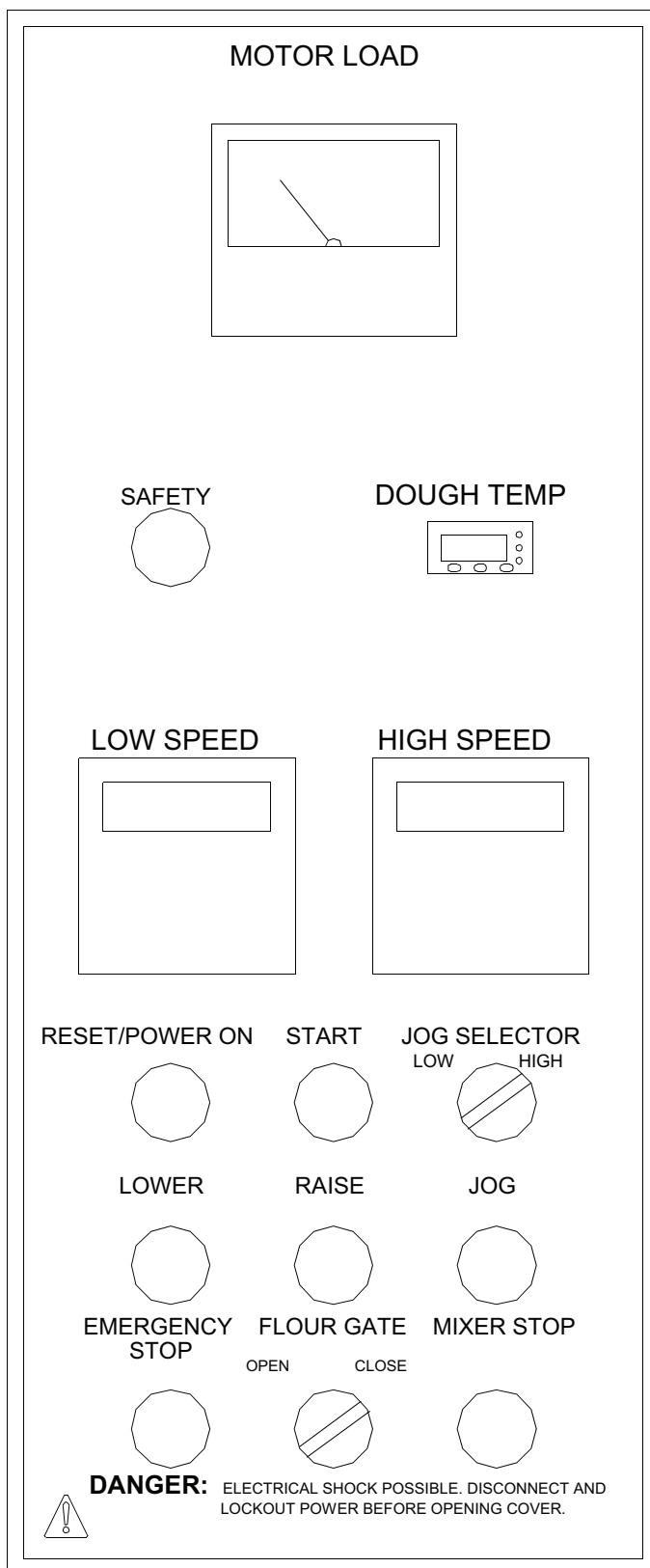
COMPONENT NAME AND TYPE	FUNCTION	INTERLOCKS
MOTOR LOAD Analog meter	Displays drive motor load as % of full load.	None
RAISE Pushbutton	Raises bowl.	Must push SAFETY button at same time. End doors, flour gate and optional safety ingredient door must be closed.
REFRIGERATION SELECTOR 3-position, illuminated selector switch	Selects coolant cycle run mode.	See Section 4-7 .
REFRIGERATION TIMER Timer	Controls time refrigeration system is on.	Starts when mixer run is started. See Section 4-7 .
RESET/POWER ON Illuminated Pushbutton	Resets timers. Stays lit when mixer power is on.	None
SAFETY Pushbutton	Used for two-hand operation when jogging, raising or lowering bowl.	Must be used with JOG , RAISE or LOWER pushbuttons.
START Pushbutton	Start mixer run cycle.	Mixer will not start if bowl is not full up, end doors are open, flour gate is open, optional safety ingredient door is open, or timers are not set.

NOTE: The **SAFETY** button must be released with the **UP**, **DOWN** or **JOG** button. If **SAFETY** button is held down by itself, the Anti-Tie Down interlock will prevent mixer from operating. Release **SAFETY** button. Repush to continue mixer operation.

NOTE: Always use **MIXER STOP** button to stop mixer in a non-emergency situation. Use **EMERGENCY STOP** button only in an emergency to reduce wear on master control relay coil and contacts. Test **EMERGENCY STOP** button at least once a week for proper operation.

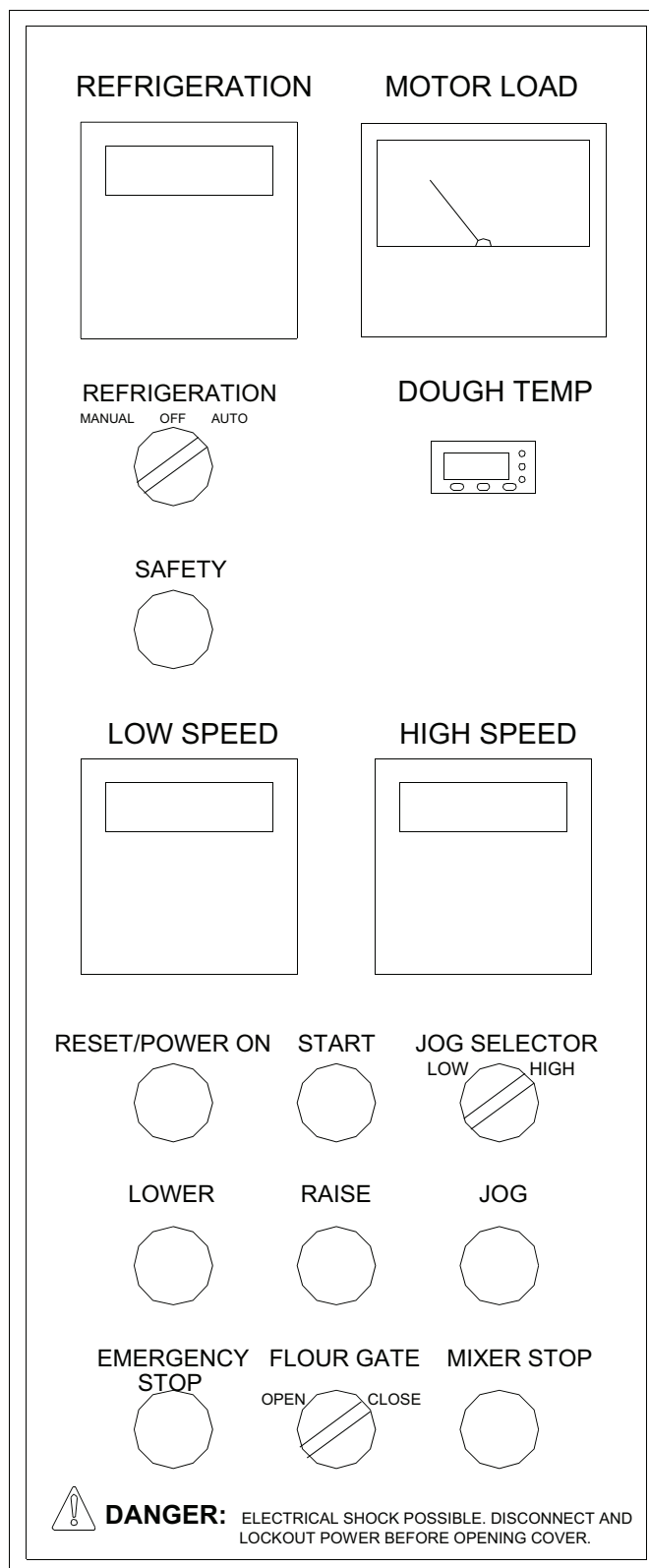
SECTION 4-1

STANDARD CONTROL PANEL FIGURE 4-1-2



SECTION 4-1

OPTIONAL CONTROL PANEL WITH REFRIGERATION FIGURE 4-1-3



SECTION 4-1

Many mixers have custom control panel layouts for special control options. See **Chapter 2, Options Checklist**, to see if your mixer has a custom panel. If your mixer is equipped with a special control panel, the function of the controls will be the same as described in this chapter. Any special or custom controls will be explained in **Chapter 8, Special Pages for Order**.

Some mixers feature a **Video Control Panel (VCP)** instead of the conventional control panel. **Figure 4-10-1** shows a typical **VCP** used on **Peerless** mixers. Other types and models of **VCPs** can be used. See **Chapter 2, Options Checklist**, for the type of control panel or **VCP** used on your mixer.

If your mixer has a **VCP**, operator controls will be the same as described in this section. Instructions for using the **VCP** are covered in **Section 4-10**. Operator controls on a **VCP** that differ from conventional control panels are described in **Section 4-10**.

The only other normal operator control that is not mounted on the control panel is the **main power disconnect**. This control is a red/black handle mounted on the right side of the electrical enclosure. Use the **main power disconnect** handle to turn the electrical power to the mixer **ON** and **OFF**.

When the handle is in the **DOWN** position, the power is **OFF**. When the handle is in the **UP** position, the power is **ON**.

Use the **main power disconnect** handle to turn the mixer's power **OFF** in an emergency, when finished operating the mixer, when cleaning the mixer, or when doing any maintenance on the mixer.

▲ DANGER

When cleaning and doing maintenance on mixer, follow **OSHA LockOut/ TagOut Standard 1910.147, The Control of Hazardous Energy**. See **Section 1-3, Zero Energy State Procedures**.

SECTION 4-2

MIX TIME, TIMER OPERATION

This section explains operation of mix timers in detail.

Peerless mixers are supplied standard with **Eagle digital timers**. They can also be supplied with optional **Eagle mechanical timers**, no timers, or a **VCP**. See **Section 2, Options Checklist**, to determine which timer type is on your mixer.

The operating sequence of the mix-cycle timers is the same for all timer types. Timer settings are explained later in this section.

Mix-cycle timers control the amount of time the mixer will operate in low and high speed. When the mixer run cycle is started, the mixer begins in low speed and runs in low speed until the **LOW SPEED** timer has timed down to zero. At this time, the mixer automatically changes to high speed and the **HIGH SPEED** timer starts to time down. When the **HIGH SPEED** timer has timed down to zero, the mixer stops.

Mix timer presets can be changed at any time. After changing presets, reset the timers by pushing the **RESET/POWER ON** button. The timers will be reset automatically for the next mix when the bowl is lowered to its full-down position and the agitator is jogged.

The mixer can be stopped anytime during the run cycle without affecting timer settings or their current timed-out value. When the cycle is restarted, the timer will continue timing where it had stopped. If the mixer was stopped in the high speed run cycle, when restarted the mixer will run in low speed for 3 to 5 seconds before changing to high speed.

SECTION 4-2**EAGLE DIGITAL TIMER OPERATION 4-2-1**

To change the setpoint of an **Eagle digital timer**, press the **SET** key located at the lower left corner of the timer. See **Figure 4-2-1, Eagle Digital Timer**. The panel keypads become active and the word **SET** appears on the bottom right corner of the screen. The setpoint is changed by pressing the ▲ or the ▼ key pads. Pressing the ▲ key increases the setpoint digit located above the keypad. Pressing the ▼ key decreases the digit located above the key pad. If either keypad is pressed and held, the digit will change rapidly until the pad is released.

When the panel displays the desired setpoint, press the **ENT** keypad and the new setpoint will be entered. Once the **ENT** keypad has been pressed, all ▲ and ▼ keypads become inoperative and **SET** disappears from the screen.

Press the **SET** pad to display or change the setpoint at any time with out affecting the current timing cycle. The current timing value is redisplayed when the **ENT** pad is pressed. When the timer is reset, the new entered value will be used.

If the mixer's power is turned off during the mix cycle, the current timing value will be kept on the timers. When power is turned back on, the mix cycle will continue where it left off.

A battery inside the timer maintains the timer setpoint when power is off to the mixer. The battery will also maintain the current timing value. With power to the mixer turned off, the battery will maintain these values for about 30 days. The battery, **Eagle Signal P/N PBB02898**, is a lithium battery with a life expectancy of 10 years.

If the setpoint is lost or a new timer installed, the timer will display four hyphens on the number screen. The timer will not operate until a setpoint has been entered.

SECTION 4-2

Select the operation of the **Eagle digital timer** using dip switches located behind the face of the timer. When replacing a timer, set the dip switches on the new timer to 1, 2, 3, 6, 7 ON and 4, 5 OFF.

EAGLE DIGITAL TIMER FIGURE 4-2-1



SECTION 4-2

EAGLE MECHANICAL TIMER OPERATION 4-2-2

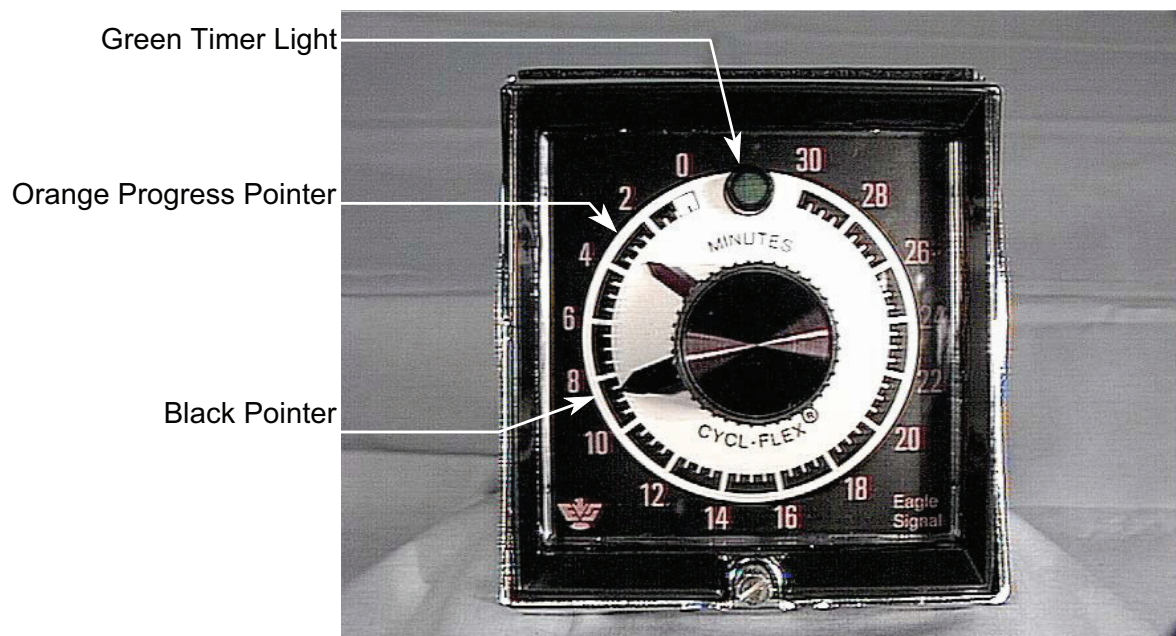
To change the setpoint of an **Eagle mechanical timer**, turn the round knob in the middle of the timer face until the black pointer points to the desired time. When the reset button is pushed, the orange cycle-progress pointer will rotate until it reaches and is stopped by the black pointer.

When the timer is timing, an green light at the top center of the timer stays lit. The red cycle-progress pointer will rotate clockwise until it reaches zero (0).

Do not use the black set pointer to force the progress pointer to zero (0). Forcing the pointer will wear out the timer clutch.

If the mixer's power is turned off during the mix cycle, the current timing value will be kept on the timers. When power is turned back on, the mix cycle will continue where it left off.

EAGLE MECHANICAL TIMER FIGURE 4-2-2



SECTION 4-3

MOTOR LOAD METER

The **MOTOR LOAD** meter is an analog meter located on the control panel of your **Peerless** mixer. See **Figure 4-1-2** or **4-1-3**. The meter measures the load on the drive motor as a percent of its rated load. For example, if the **MOTOR LOAD** meter reads 90, the drive motor is pulling 90% of its rated capacity.

It is normal for the **MOTOR LOAD** meter to peak at 150-200% for very short periods of time. However, if the meter runs at greater than 100% for more than 5 seconds, the drive motor may be overloaded. If the **MOTOR LOAD** meter consistently shows a possible overload, call maintenance or the Service Department at **Peerless Machinery Corp.**

The **MOTOR LOAD** meter can indicate when a problem with the ingredient loading has occurred. A reading higher or lower than normal could mean the proper ingredients have not been loaded.

SECTION 4-4

DOUGH THERMOMETER

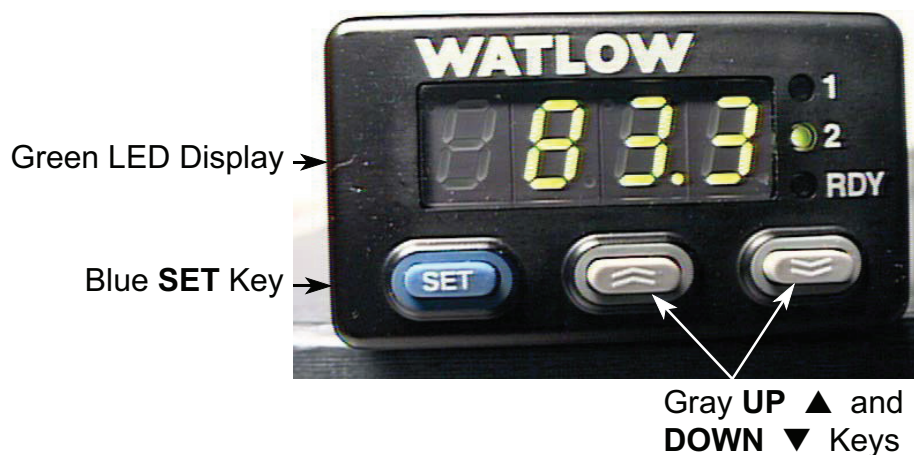
Your **Peerless** mixer comes standard with a digital thermometer. The **dough thermometer** measures the dough temperature in the bowl. It is a display only and does not control dough temperature.

Due to inconsistencies of dough contact with the sensor as well as interference of the bowl refrigeration jacket, the actual readout may be different from the temperature reading of an independent thermometer.

The **dough thermometer** can be set to display in either degrees Celsius or Fahrenheit. To change the display units, see the **Change Unit of Measure** procedure in this section.

The **dough thermometer** display value can be adjusted to match the actual value. Adjust the display value by entering the desired change in the displayed value (offset) in the thermometer. See the **Display Offset** procedure below. Refer to **Figure 4-4-1, Dough Thermometer**.

DOUGH THERMOMETER FIGURE 4-4-1



SECTION 4-4**SOFTWARE LOCKOUT:**

The **dough thermometer** is preset at the factory and locked out with the exception of the **tAg** setting. To view or change the program settings, change the **tAg** settings to a blank readout using the following procedure. Refer to **Figure 4-4-1, Dough Thermometer**.

1. Press and *hold* the gray **UP ▲** and **DOWN ▼** keys for 3 seconds to go into the program mode. The display will show **CnFg**.
2. Press and *hold* the blue **SET** key and at the same time press the **DOWN ▼** key until you see **YES**. Release the **DOWN ▼** key and **SET** key. If the lockout option is set, **tAg** will be displayed.
3. Press and *hold* the **SET** key. **PCOA** will be displayed. While holding down the **SET** key, push the **DOWN ▼** key several times until the display is blank. Release the **DOWN ▼** key and **SET** key.

NOTE: *The dough thermometer is now unlocked.* Make any changes to the program now, before continuing with *steps 4-6* which tell how to **lock** the program.

4. After the program has been modified as desired, reset the lockout. Cycle through the program using the **DOWN ▼** key until **tAg** is displayed. (**tAg** is under the **CnFg** menu.)
5. Press and *hold* the **SET** key, and at the same time push the **DOWN ▼** key several times until the display reads **PCOA**. Release the **SET** key.
6. The temperature value will appear after 60 seconds.

DISPLAY OFFSET:

1. Remove software lockout per above **Software Lockout** procedure.
2. Press and *hold* both the **UP ▲** and **DOWN ▼** keys for 3 seconds to go into the program mode.

SECTION 4-4

3. Press the **DOWN ▼** key until you see the **Pid** prompt. Press and *hold* the **SET** key, and at the same time push the **DOWN ▼** key until you see **YES**. Release the **DOWN ▼** key and **SET** key.
4. Press the **DOWN ▼** key until you see **CAL**.
5.
 - a) To raise the display value: Press and *hold* the **SET** key and at the same time push the **UP ▲** key until you reach the number of degrees you want to raise the display value. Release the **SET** key and **UP ▲** key.
 - b) To lower the display value: Press and *hold* the **SET** key, and at the same time press the **DOWN ▼** key until you reach the number of degrees you want to lower the display value. Release the **SET** key and **DOWN ▼** key.
6. The new temperature value will appear after 60 seconds.

CHANGE UNIT OF MEASURE:

1. Remove software lockout per above **Software Lockout** procedure.
2. Press and *hold* both the **UP ▲** and **DOWN ▼** keys for 3 seconds to go into the program mode.
3. Press the **DOWN ▼** key until you see **CnFg** displayed.
4. Press and *hold* the **SET** key, and at the same time press the **DOWN ▼** key until you see **YES**. Release the **DOWN ▼** key and **SET** key.
5. On the display screen, **In** will be displayed. Push the **DOWN ▼** key until **C_F** is displayed.
6. Press and *hold* the **SET** key. The letter displayed is the current unit of measure of the thermometer. *F* means the display is in Fahrenheit, and *C* means the display is in Celsius. Push the **DOWN ▼** key to change to a different display unit. When the desired display unit is shown, release the **DOWN ▼** key and **SET** key.
7. The temperature value will appear in the selected unit after 60 seconds.

SECTION 4-4

DEFAULT PROGRAM SETTINGS:

If the **dough thermometer** doesn't work or you have a new unit, set the program according to the default **CnFg** settings below. See the previous section for how to program the **dough thermometer**.

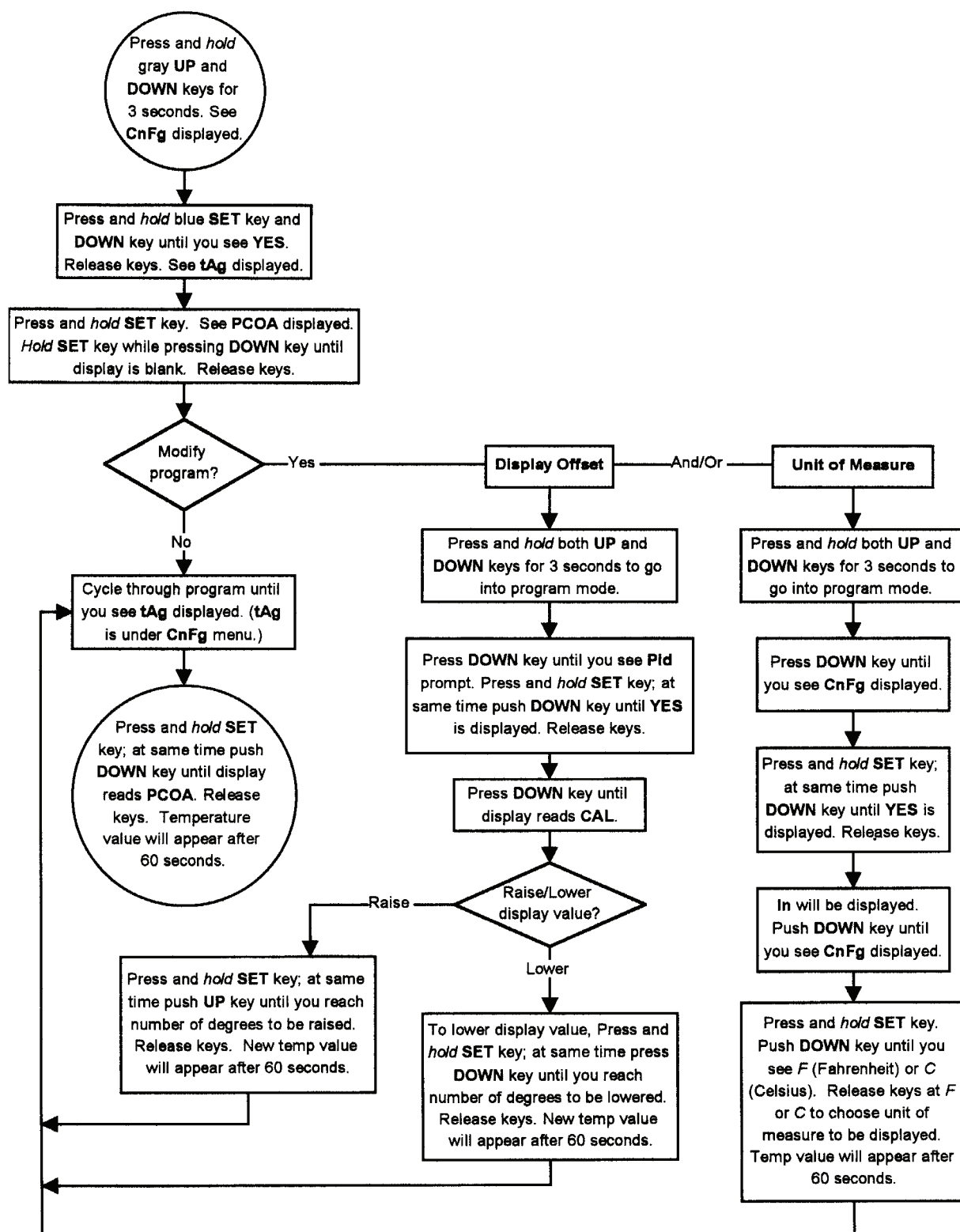
CnFg Menu	
Displayed	Selected
ln	rt.d
C_F	F
rL	-99.9
rh	999.9
Ot1	nonE
Ot2	COOL
diSP	Act (Ac)
FAIL	bPLS
SLOC	no
tAg	PCOA

If the display is in celsius, for the **C_F** value, select **C**.

In the **Pid** menu, press the **DOWN ▼** key repeatedly until you see **CAL** and set the calibration following steps 5 and 6 of the **Display Offset** procedure above. The settings in **Aut** (Autotune) menus do not matter in display-only **dough thermometer** units.

SECTION 4-4

SOFTWARE LOCKOUT PROCEDURE FLOW CHART 4-4-2



SECTION 4-5

DOUGH TEMPERATURE CONTROLLER

An optional **dough temperature controller** measures and displays the actual dough temperature in the bowl, and uses the dough temperature to control the refrigeration system. The thermometer can be set to display either degrees Celsius or Fahrenheit. See **Section 4-4, Dough Thermometer**, for the procedure on how to change the display units.

The refrigeration system is controlled by a setpoint entered in the **dough temperature controller**. As long as the dough temperature stays below the setpoint, the system will not refrigerate. If the dough temperature rises above the setpoint, the refrigeration unit will operate to maintain the desired dough temperature.

SETPOINT ADJUSTMENT PROCEDURE

1. Press and *hold* the blue **SET** key. The current setpoint value will be displayed.
2. While holding the **SET** key down, press either the **UP ▲** or **DOWN ▼** key until the desired setpoint value is reached.
3. Once the desired setpoint value is reached, release the **SET** key.

See **Section 4-7** for the procedure on operating the refrigeration system using a **dough temperature controller**.

Due to inconsistencies of dough contact with the sensor as well as interference of the bowl refrigeration jacket, the actual readout may be different from the temperature taken with an independent thermometer. The thermometer display value can be adjusted to match the actual value. Adjust the display value by entering the desired change in the displayed value (offset) in the thermometer. See the **Display Offset** procedure in **Section 4-4, Dough Thermometer**.

SECTION 4-6

OPERATING PROCEDURES

⚠ DANGER

Having more than one mixer operator at the same time creates a hazardous situation which can result in serious injury or death. *Never reach into the bowl while someone else is next to the mixer's operator panel.*

⚠ DANGER

Running or coasting mixing arm can cause serious injury or death. Never reach hand or arm inside bowl while mixing arm is moving.

⚠ DANGER

Moving dough can cause serious injury. Never reach into bowl through optional **ingredient door** while dough or agitator is moving.

⚠ DANGER

Running or coasting mixing arm can cause serious injury or death. Never place upper body or head inside mixing bowl without disconnecting and locking out power.

⚠ CAUTION

Before operating this machine, read, understand and follow **Chapter 1, Safety**, and **Chapter 4, Operations, Section 4-1 through 4-6**.

NOTE:

Peerless mixers have an "anti-tie down feature." If **SAFETY** button alone is held down for more than two seconds, the mixer will become inoperable. Bowl will not move, and agitator will not jog. To reset, release **SAFETY** button.

This two-part section covers the operation procedure for your **Peerless** mixer. **Section 4-6-1, Simplified Operating Procedure**, gives basic step by step instructions for operating your mixer. **Section 4-6-2, Detailed Operating Procedure**, gives a detailed description of operation and outlines various ways your mixer can be operated.

SECTION 4-6

The safety systems and features on **Peerless** mixers are designed and installed to protect one mixer operator only. Using a helper to assist the operator in discharging dough is not considered safe and must never be done. *Never reach into the bowl when someone else is next to the mixer's operator panel.*

Some **Peerless** mixers have special operational procedures that will not be covered in this section. Any special procedures will be covered in **Chapter 8, Special Pages for Order**. See **Chapter 2, Options Checklist**, for mixer features that may require special procedures.

See **Section 4-1** for description and detail of all operator controls named in the following sections.

SIMPLIFIED OPERATING PROCEDURE 4-6-1

See **Figure 4-6-1, Operating Procedure Flow Chart**, for typical operation.

NOTE: Use **EMERGENCY STOP** button to stop mixer only in an emergency.

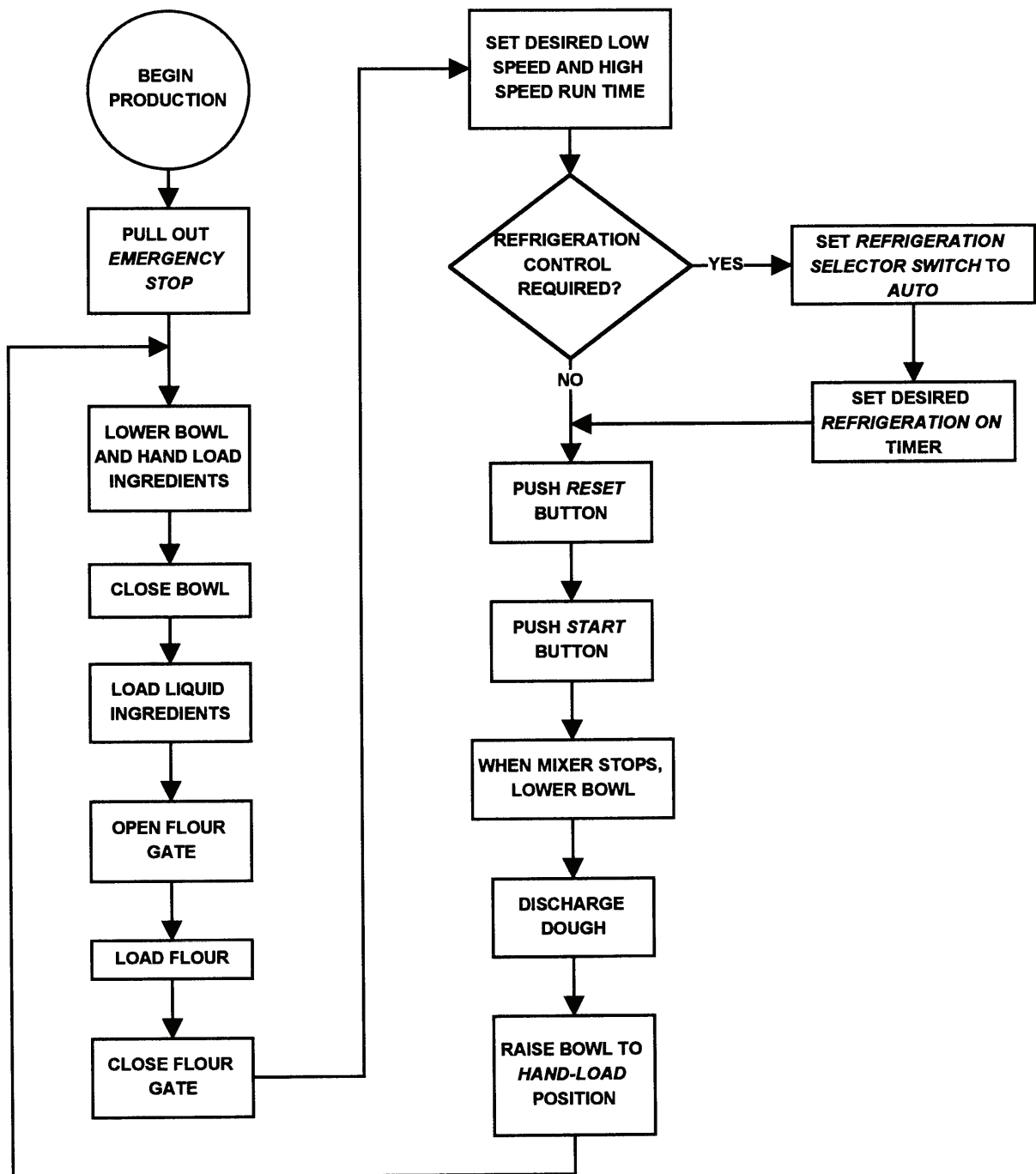
NOTE: For **RAISE**, **LOWER** and **JOG** buttons to work, **SAFETY** button must be pushed at same time.

NOTE: When first starting, mixer will always run for three seconds in low speed before going into high speed.

NOTE: Before mixer will start, **EMERGENCY STOP** button must be pulled out.

SECTION 4-6

TYPICAL OPERATING PROCEDURE FLOW CHART 4-6-1



SECTION 4-6

The following is a typical operator procedure for a standard **Peerless** mixer.

1. If necessary, tilt the bowl down to hand-load ingredients.
2. Load all liquid ingredients as required.
3. Open the **flour gate** before loading flour by turning the **FLOUR GATE** selector switch to OPEN.
4. After loading flour, close the **flour gate** by turning the **FLOUR GATE** selector switch to CLOSE.
5. Set desired mix time on the **LOW SPEED** and **HIGH SPEED** timers.
6. (Optional) Set desired refrigeration on time on the **REFRIGERATION** timer and also set the **REFRIGERATION** selector switch to **AUTO** position.
7. Push the **RESET/POWER ON** lighted button to reset timers.
8. Push the **START** button to start the mix cycle.
9. The mixer can be stopped at any time by pushing the **MIXER STOP** button. To restart the mixer, push **START** button.
10. When the mix cycle is over, lower the bowl by pushing and holding the **LOWER** and **SAFETY** buttons. The bowl will stop when it reaches its full down position.
11. Select jog speed by setting the **JOG SELECTOR** switch to either **LOW** or **HIGH** speed.
12. Discharge dough by pushing and holding both **JOG** and **SAFETY** buttons until the dough has discharged.

▲ DANGER

Having more than one mixer operator at the same time creates a hazardous situation which can result in serious injury or death. *Never reach into the bowl while someone else is next to the mixer's operator panel.*

▲ DANGER

Running or coasting mixing arm can cause serious injury or death. Never reach hand or arm inside bowl while mixing arm is moving.

SECTION 4-6

**CAUTION**

Do not cycle drive motor on and off repeatedly in a short time. Damage to motor and motor starter can result.

13. Raise the bowl by pushing and holding the **RAISE** and **SAFETY** buttons. The bowl will stop when it reaches its full up position.
14. Start over at step 1 for additional batches.

DETAILED OPERATING PROCEDURE 4-6-2

The following is a detailed description of the operation of a standard **Peerless** mixer. Some common options have been included. See **Chapter 2, Options Checklist**, to find out which options, if any, are included on your mixer.

NOTE: Use **EMERGENCY STOP** button to stop mixer only in an emergency.

NOTE: For **RAISE**, **LOWER** and **JOG** buttons to work, **SAFETY** button must be pushed at same time.

NOTE: On starting or restarting, mixer will always run for three seconds in low speed before going into high speed.

NOTE: Before mixer will start, **EMERGENCY STOP** button must be pulled out.

INGREDIENT LOADING

Load ingredients into mixer by one or more of the following methods:

- ✓ Hand loading from front of mixer with bowl open part way.

Hand-load ingredients from the front of the mixer by lowering the bowl part way and placing ingredients in front of the bowl. The bowl will stop in any position when you release either the **SAFETY** or **LOWER** button.

SECTION 4-6

- ✓ Hand loading from rear through optional **safety ingredient door**.

An optional **safety ingredient door** can be supplied so ingredients can be loaded from the rear of the mixer. Normally a platform is used at the rear of the mixer so the **safety ingredient door** can be reached. Do not use a ladder to carry ingredients up to door. This practice is unsafe, and could cause injury from a fall.

The mixer will not run when the **safety ingredient door** is open. If the door is opened while the mixer is running, the mixer will stop. To restart the mixer, close and latch the door; push the **START** button.

On some sizes of **safety ingredient doors**, a safety grid has been welded in place for your protection. **Do not remove safety grid.**

- ✓ Loading from a bulk flour system through **flour gate** opening.

Load bulk flour into the bowl through the flour inlet on top of the mixer. The standard inlet is a sliding gate that opens or closes the inlet. An optional **butterfly valve** can also be supplied.

Open the flour inlet using the green **FLOUR GATE** selector switch. The **FLOUR GATE** selector switch will light up when OPEN is selected, and the gate will open. Turn the selector switch to CLOSE to close. The **flour gate** will not open if the mixer is running or the bowl is down.

The mixer will not tilt or run if the inlet is open. After opening the mixer's flour inlet, the valve on the bulk flour hopper should also be opened. See the operator's manual on bulk flour systems for instructions.

- ✓ Loading from a liquid system or systems through liquid inlets located on top of mixer.

Liquid can be loaded through openings made in the mixer according to customer needs. Typically, openings are located on the top front of the bowl canopy.

SECTION 4-6

TIMER SETTING, MIXER STARTING AND STOPPING

1. Set timers. See **Section 4-2** for setting mixer run-cycle timers.
2. Once run timers are set, push the **RESET/POWER ON** button. Run cycle timers will reset.

NOTE: Before mixer will start, **EMERGENCY STOP** push/pull button must be pulled out.

3. Stop the mixer at any time by pushing the **STOP** button. With the mixer stopped, tilt the bowl or open the optional **safety ingredient door** to hand load additional ingredients if required.
4. To restart the mixer, push the **START** button. The mixer will continue the run cycle where it left off. If the mixer was stopped while running in high speed, it will restart in low speed for 3 seconds before changing to high speed.
5. When run-cycle timers have timed out to zero (0), the mixer will stop.

DISCHARGING DOUGH

Discharge dough into a dough trough, through the floor into a conveyor discharge hopper, or onto a conveyor. Hold firmly in place any device the dough is being discharged into or onto. For dough troughs, an optional **trough hook** can be used. Lift the **trough hook** and place inside the trough before the dough is discharged. The **trough hook** will hold the trough stationary while the dough loads into it.

 **CAUTION** Discharging dough can cause trough to roll away from mixer causing personal injury. Do not stand in front of trough when discharging dough.

SECTION 4-6

Use the following method to discharge dough:

1. To lower the bowl, push and hold both **LOWER** and **SAFETY** buttons. The bowl will stop automatically in full down position.

NOTE: If **JOG** button is pushed while bowl is being lowered, bowl will stop, agitator will start, then bowl will restart.

2. Jog the agitator to help kick dough out of the bowl. Push and hold both **JOG** and **SAFETY** buttons. Jog the agitator in either high speed or low speed by using the **JOG SELECTOR** two-position selector switch. Switch to the desired jog speed.

⚠ DANGER Having more than one mixer operator at the same time creates a hazardous situation which can result in serious injury or death. *Never reach into the bowl while someone else is next to the mixer's operator panel.*

⚠ DANGER Running or coasting mixing arm can cause serious injury or death. Never reach hand or arm inside bowl while mixing arm is moving.

3. To discharge sticky dough, start to jog the agitator in high speed before the **HIGH SPEED** timer times out. Select high-speed jog with the **JOG SELECTOR** switch and push the **JOG** and **SAFETY** buttons while a few seconds still remain on the **HIGH SPEED** timer. Push and hold the **LOWER** button. This technique keeps dough from falling into the agitator.

⚠ CAUTION: Do not cycle drive motor on and off repeatedly in a short time. Damage to motor and motor starter can result.

4. After discharge of dough, raise the bowl by pushing and holding the **RAISE** and **SAFETY** buttons.

SECTION 4-6

JOG-AND-TILT

It may be easier to discharge some doughs by taking advantage of the **Jog-and-Tilt** feature of your **Peerless** mixer. With this feature, you can lower the bowl while the agitator is jogging. Push and hold the **JOG**, **LOWER** and **SAFETY** buttons at the same time. The agitator must be running before the bowl can be lowered.

Peerless can automate just about any mix cycle. It is best to specify any special mix cycle functions at the time of order. However, it is possible to modify the mixer run cycle for some special mix functions after your mixer has been installed.

NOTE: If you have any questions, consult the Engineering Department at Peerless Machinery Corp.

SECTION 4-7

REFRIGERATION OPERATION PROCEDURE

A **refrigeration control circuit**, optional on Double Arm mixers, controls the refrigeration system. See **Chapter 2, Options Checklist**, to see if your mixer has a **refrigeration control circuit**.

A lighted, three-position selector switch, labeled **REFRIGERATION**, controls your mixer's refrigeration system. The **REFRIGERATION** switch stays lit whenever the system is operating and can be set at **OFF**, **MANUAL** or **AUTO**.

If the **REFRIGERATION** selector switch is in the **OFF** position, the refrigeration system remains off. No refrigerant flows to the mixer.

If the **REFRIGERATION** selector switch is in the **MANUAL** position, the refrigeration system stays on. Refrigerant flows continuously to the mixer.

If the **REFRIGERATION** selector switch is in the **AUTO** position, the refrigeration timer controls the refrigeration system. Use the following procedure.

1. Set desired refrigeration on-time on the **REFRIGERATION** timer. See **Section 4-2** for instruction on setting timers. The refrigeration on-time required is determined by experience based on observation of refrigeration on-time and the final dough temperature of the previous batch.
2. Set the **REFRIGERATION** selector switch to **AUTO** position.
3. Push the **RESET** button.
4. Push the mixer **START** button.

When the mixer **START** button is pushed, the mix cycle and refrigeration system start. Refrigerant begins to flow through the bowl jacket. The **REFRIGERATION** timer starts timing down.

When the **REFRIGERATION** timer reaches zero (0), the refrigeration system shuts down and refrigerant stops flowing through the bowl jacket.

When the mix timers are reset, the **REFRIGERATION** timer automatically resets.

SECTION 4-8

OPERATOR TROUBLE-SHOOTING GUIDE

- NOTE:**
- 1) Call maintenance for any problems not covered here.
 - 2) To use **RAISE**, **LOWER** AND **JOG** buttons, push **SAFETY** button at same time.
 - 3) To operate mixer, you must pull out **EMERGENCY STOP** button.

OPERATOR TROUBLE-SHOOTING GUIDE TABLE 4-8-1

TROUBLE	CAUSE	REMEDY
Mixer will not start.	Bowl not fully closed.	Close bowl.
	Timer set wrong.	Reset or enter time.
	Safety Ingredient door open.	Close and latch safety ingredient door.
	End door open.	Close end door.
	Flour gate open.	Close flour gate.
Bowl will not tilt.	Safety ingredient door open.	Close and latch safety ingredient door.
	End door open.	Close end door.
	Flour gate open.	Close flour gate.
Agitator will not jog.	Safety ingredient door open.	Close and latch safety ingredient door.
	End door open.	Close end door.
	Flour gate open.	Close flour gate.
Refrigeration will not run.	REFRIGERATION selector switch in OFF mode.	See Section 4-7 or call maintenance.
Flour gate will not open.	Bowl not closed.	Close bowl.
	No air pressure.	Call maintenance.

SECTION 4-9

OPTIONAL EQUIPMENT OPERATION

This section explains the operation of some of the optional features which may be found on your **Peerless** mixer. See **Chapter 2, Options Checklist**, to find out if your mixer has any of these options.

CYCLE-OVER HORN

The **cycle-over horn** sounds when the **HIGH SPEED** timer times out and signals the end of the cycle.

TROUGH HOOKS (TRO HOOKS)

The **trough hooks**, which are attached to the mixer, hook onto the trough to keep it from rolling away during dough discharge. There are two types of **trough hooks**: **manual** and **pneumatic**.

If your mixer has **pneumatic trough hooks**, there is a green selector switch labeled **TRO HOOK** on the operator's panel. Turn the **TRO HOOK** selector switch to LOCK to lower the **pneumatic trough hooks** and latch the trough. Select UNLOCK to raise the **trough hooks** and unlatch the trough.

SAFETY INGREDIENT DOOR

DANGER Running or coasting mixing arm can cause serious injury or death. Never reach hand or arm inside bowl while mixing arm is moving.

 **WARNING** A falling **safety ingredient door** can cause severe personal injury.

NOTE: Mixer will not run if **safety ingredient door** is open.

There are two types of **safety ingredient doors**: **manual swing** and **pneumatic sliding**. **Safety ingredient doors** allow ingredient loading from the top or rear of the mixer.

Pneumatic safety ingredient doors have a push/pull button located next to the door.

HAND-LOAD POSITION

In the **hand-load position**, the bowl automatically stops so the operator can add ingredients like salt, yeast, and sugar by hand.

After discharging the dough and starting the bowl back up, the bowl will stop so the operator can add ingredients. After the bowl stops, the **SAFETY** and **UP** buttons must be released, then pushed again, for the bowl to continue to close.

SECTION 4-10

VIDEO CONTROL PANELS

Your **Peerless** mixer may be equipped with an optional **Video Control Panel (VCP)**. See **Chapter 2, Options Checklist**, to find out if your mixer has a **VCP**. Video controllers are used in place of pushbutton control panels. This section covers the three standard types of **VCPs** used by **Peerless**. They are: **Cutler-Hammer IDT PanelMate® 1000**, **Cutler-Hammer IDT PanelMate® 3000**, and **Allen-Bradley PanelView® 550**. Other types of **VCPs** can be used at customer request; see **Chapter 8, Special Pages for Order**. This chapter gives an overview of each of the above mentioned models.

Many of the buttons and controls are the same on **VCPs** as on standard pushbutton control panels. For a summary of the standard controls and buttons, see Chart **4-1-1, Operator Controls Summary Chart**, in **Section 4-1**. Controls and buttons unique to **VCPs** will be covered in **Section 4-10-5**.

The operating procedure of a mixer with a **VCP** is the same as a mixer with a standard pushbutton control panel. See **Section 4-6** for the standard mixer operating procedure. If your mixer has refrigeration control, see **Section 4-7**.

If your mixer has been equipped with a **PanelMate® 3000**, a **Multiple Product Mixer Control Feature (MPMCF)** is supplied, allowing for storage of mixing presets for various product types. See **Section 4-10-4** for instructions on how to use the **MPMCF**.

Customers commonly request special operating procedures using **VCPs**. See **Chapter 2, Options Checklist**, to find out if your mixer has special operating procedures. If so, the special operating procedure will be listed in **Chapter 8, Special Pages for Order**.

NOTE: **MIX CYCLE CODES** and **AGITATOR HOUR CLOCK** values are not stored in permanent memory or in the **EEPROM** memory. This information is stored in the **PLC** database. If it becomes necessary to revise the PLC program, upload the program before making the necessary changes to the program to assure that the original **PLC** database will not be lost.

SECTION 4-10

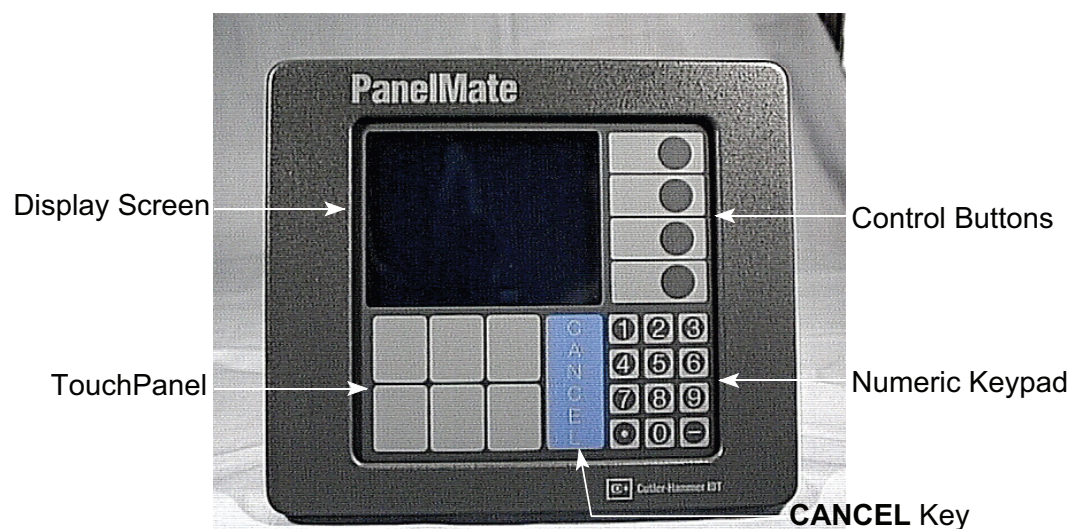
CUTLER-HAMMER IDT PANELMATE® SERIES 1000 4-10-1

Your **Peerless** mixer can be equipped with an optional **Cutler-Hammer IDT PanelMate®1000 Series Video Control Panel (VCP)** to provide mixer control. This section covers basic operation of the **IDT PanelMate 1000**.

The **Eaton IDT PanelMate 1000 VCP** is a compact operator-interface device for use with a **Programmable Logic Controller (PLC)**. A **VCP** takes the place of typical control panel components like indicator lights, pushbuttons, timers and digital displays like thermometers. When mounted on the mixer, the **PanelMate 1000** is completely sealed against dust, dirt, oil and moisture. It has a NEMA 4X rating and will withstand moderate wash down with low-pressure water.

The front panel of the **PanelMate 1000** unit is shown below. The 5" (12.7cm) display area displays the graphic templates that control mixer operations.

CUTLER-HAMMER IDT PANELMATE® 1000 VCP FIGURE 4-10-1

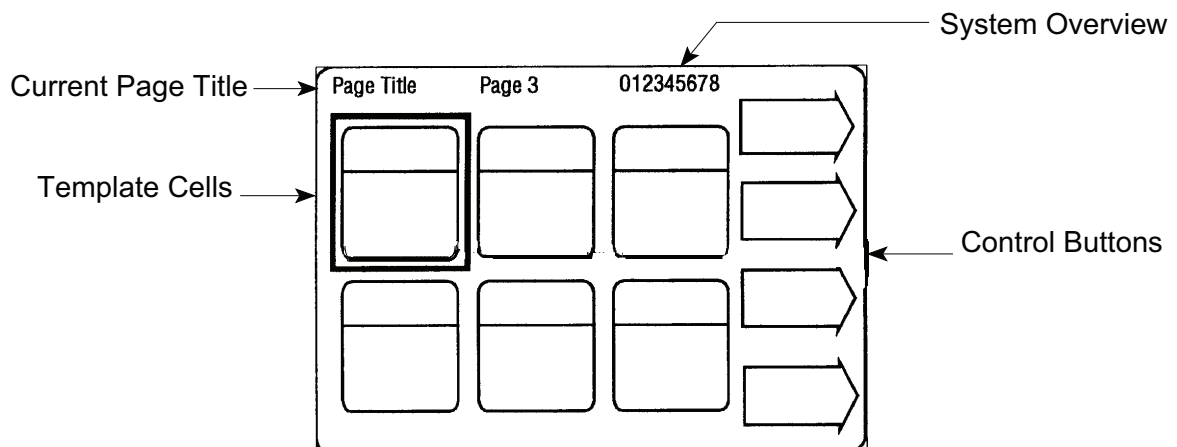


When power is turned on to your mixer, the **Cutler-Hammer IDT PanelMate 1000** unit will perform various system diagnostics on itself to make sure it is working properly. If the system encounters a problem, turn the power off and on again to see if the problem is repeated. If the **VCP** reports the same problem, call your local **Cutler-Hammer** distributor or the Engineering Department at **Peerless Machinery Corp.**

After the self-diagnostics check is run, page zero (0) of the mixer control software package will be displayed. See **Figure 4-10-2, General Page Layout**, below.

CUTLER-HAMMER IDT PANELMATE® 1000 GENERAL PAGE LAYOUT

FIGURE 4-10-2



SELECT TEMPLATE CELLS

The six squares on the **TouchPanel** correspond to the six **template cells** on the display screen. Some **template cells** take up to three cell squares. Select a **template cell** by firmly touching the area on the **TouchPanel** that corresponds to the location of the template on the screen. When a **template cell** has been selected, the **VCP** will indicate that the cell is active by displaying a highlighted square on the screen around the selected **template cell**.

SECTION 4-10

A small dark arrow at the bottom of a **template cell** means the cell has at least one **control button** function assigned to it. When one of these **template cells** is selected, **control button labels** will appear to the right of the screen next to the **control button** used for it. Pressing the **control button** will activate the control function.

For example, a **template cell** on the screen might be labeled **MIXER CONTROLS**. Firmly touch the square on the **TouchPanel** that corresponds to the **MIXER CONTROLS template cell**. The **control button labels** assigned to the **MIXER CONTROLS template cell** will appear at the right of the screen pointing to corresponding **control buttons**. In addition, you will see a bright square around the **template cell MIXER CONTROLS**. If the top **control button label** displays **START CYCLE**, pressing the top **control button** will start the mixer run cycle if all safety and interlock conditions have been met.

CHANGE DISPLAY PAGE NUMBER

The **IDT PanelMate 1000** can be programmed to display up to eight different screens, or pages, on the **page display screen**. The pages are labeled starting with page zero (0) and ending with page seven. Each page displays specific information and control commands for the mixer.

The default page that appears on powering up of the unit is page zero (0). To "turn the page," push the **control button** next to the **control button label GET PAGE**. If a **template cell** has already been selected, touch the blue **CANCEL** key on the **TouchPanel**. This action will deselect the previously selected **template cell** and display the default **GET PAGE control button label**.

After the **GET PAGE control button** is pushed, the **VCP** will display an **ENTER PAGE # control button label**. At this point, touch the number of the desired page on the **numeric keypad** on the **TouchPanel**. The currently displayed page will be replaced with the page number entered.

SECTION 4-10

ENTER OR CHANGE NUMERIC VALUE

For most applications of the **IDT PanelMate 1000**, timer values for the mixer run cycle must be entered. Enter numeric values by selecting the **template cell** for the value to be changed. When the **template cell** is activated, the screen will display the **control button label CHANGE VALUE**. Touch this **control button** to change the timer's numeric value. After the **CHANGE VALUE control button** is touched, the **control button label ENTER** will be displayed. Enter the new number using the **numeric keypad** on the **TouchPanel**. Be sure to enter decimal points as needed.

NOTE: All time values are entered in hundredths of a minute--not in minutes and seconds. Enter 1 minute and 45 seconds as 1.75 minutes.

The value entered will be shown on the **control button label** next to the **bottom control button**. If this value is correct, touch the **ENTER control button** to enter the value into the system.

ADJUST SCREEN CONTRAST

The **VCP** screen contrast can be changed anytime. To change, push the **GET PAGE control button**. The top **control button** will have the **control button label ADJUST CONTRAST** displayed beside it. After pushing this button, the **control button labels LIGHTEN** and **DARKEN** will be displayed beside the top two **control buttons**.

Push the **DARKEN control button** to darken the screen or increase screen contrast. Push the **LIGHTEN control button** to lighten the screen or decrease screen contrast. Push and hold the **DARKEN** or **LIGHTEN** button until on-screen template and **control button** information can be easily seen.

After reaching the desired screen contrast, push the **SAVE control button** to save the contrast setting.

SECTION 4-10

AUTOMATIC PAGE CHANGE FEATURE

Your mixer's **IDT PanelMate 1000** has been programmed with an optional **Automatic Page Change Feature**. This feature will allow the pages to change automatically when the mixer is placed in a run cycle or when it is stopped.

The **Automatic Page Change Feature** makes it possible to display functions and information needed to control the mixer in various operation modes on the screen automatically. When this option is turned on, the **PanelMate** screen will change from the default stopped screen to the default running screen when the **CYCLE START control button** is pushed. When the mixing cycle is over or any time the mixer is stopped, the screen will change from the default running screen to the default stopped screen.

NOTE: For some versions of **IDT PanelMate 1000** software, screens will only change automatically if no **template cell** is selected or active. If a **template cell** is selected, touch the blue **CANCEL** key on the **TouchPanel**. For example, when the **CYCLE START control button** is active, you must press the **CANCEL** key before the screen will change.

To enable your mixer's **Automatic Page Change Feature**, values and settings on the screen control page must be set. For the standard mixer control package, the screen control page number is six (6).

The screen control page will display **template cells** called **AUTO SCR CHANGE**, **STOPPED SCREEN #**, and **RUNNING SCREEN #**. To enable **Automatic Page Change**, select the **template cell AUTO SCR CHANGE** and push the **YES** control button. Then select the **STOPPED SCREEN # template cell** and enter the number of the default page. This will usually be page zero (0). Next, select the **RUNNING SCREEN # template cell** and enter the number of the default running page. **Automatic Page Change** or the default page numbers can be turned off or changed at any time.

SECTION 4-10

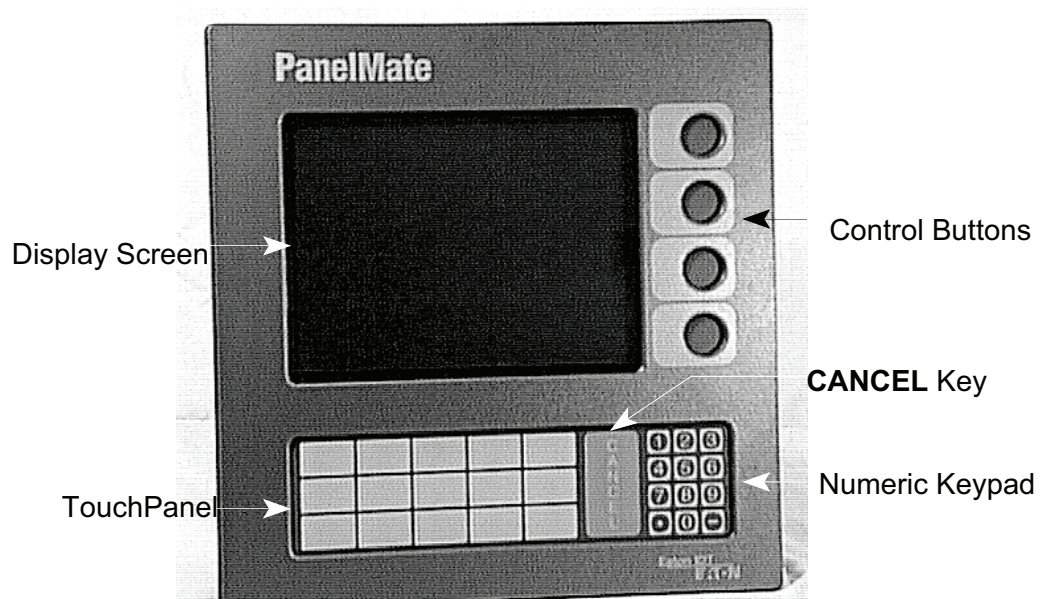
CUTLER-HAMMER IDT PANELMATE® POWER SERIES 3000 4-10-2

Your mixer can be equipped with an optional **Cutler-Hammer IDT PanelMate® 3000 Power Series Video Control Panel (VCP)** to provide mixer control. This section covers the basic operation of the **IDT PanelMate 3000 VCP**.

The **IDT PanelMate 3000 VCP** is a compact operator-interface device for use with **PLCs**. The **VCP** takes the place of typical control panel components like indicator lights, pushbuttons, timers and digital displays like thermometers. When mounted on the mixer, the **PanelMate 3000 VCP** is completely sealed against dust, dirt, oil and moisture. The **PanelMate 3000 VCP** has a NEMA 4X rating and will withstand moderate wash down with low-pressure water.

See the front panel of the **PanelMate 3000 VCP** unit below. The 9.4" (24 cm.) screen displays a variety of graphic templates that replaces the functions of traditional hand-wired operator devices such as pushbuttons, lamps and message/numeric displays. **Peerless** can supply the display in either gray scale or color.

CUTLER-HAMMER IDT PanelMate 3000® VCP FIGURE 4-10-3

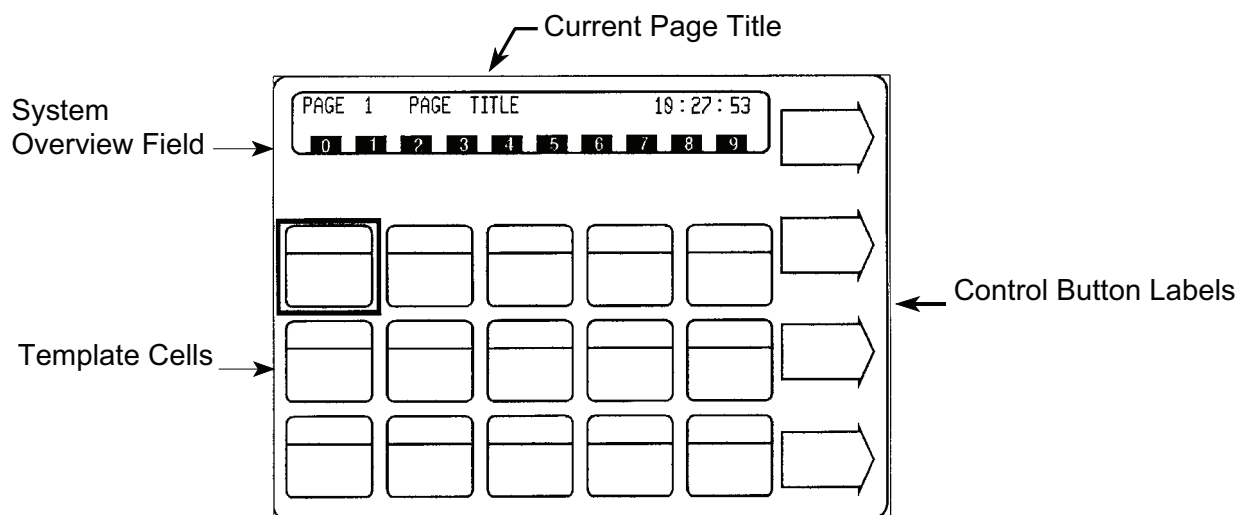


SECTION 4-10

When your mixer's power is turned on, the **PanelMate 3000 VCP** unit will perform various system diagnostics on itself to make sure it is working properly. If a problem is encountered, turn the power off and on again to see if the problem is repeated. If the **VCP** reports the same problem, call your local Cutler-Hammer distributor or the Engineering Department at **Peerless Machinery Corp.**

After the self-diagnostics check is complete, page zero (00) of the mixer control software package as well as the current time will be displayed. See **Figure 4-10-4, PanelMate 3000 VCP General Page Layout**, below.

PanelMate 3000® VCP GENERAL PAGE LAYOUT FIGURE 4-10-4



SELECT TEMPLATE CELLS

The fifteen squares on the **TouchPanel** correspond to the fifteen **template cells** on the display screen. Some **template cells** take up to four cell squares. Select a **template cell** by pushing the area on the **TouchPanel** that corresponds to the location of the template on the screen. A bright square will surround the selected **template cell** to show the **template cell** is active.

SECTION 4-10

A small dark arrow at the bottom of a **template cell** means the cell has at least one **control button** function assigned to it. When one of these **template cells** is selected, **control button labels** will appear to the right of the screen next to the **control button** used for it. Pressing the **control button** will activate the control function.

For example, a **template cell** on the screen might be labeled **MIXER CONTROLS**. Firmly touch the square on the **TouchPanel** that corresponds to the **MIXER CONTROLS template cell**. The **control button labels** assigned to the **MIXER CONTROLS template cell** will appear at the right of the screen pointing to the corresponding **control buttons**. In addition, a highlighted square will surround the **MIXER CONTROLS template cell**. If the top **control button label** says **START CYCLE**, pressing the top **control button** will start the mixer run cycle--if all safety and interlock conditions have been satisfied.

CHANGE DISPLAY PAGE NUMBER

The **PanelMate 3000 VCP** can be programmed to display up to fifteen different screens, or pages, on the graphic display screen. The pages are labeled starting with page zero (00) and ending with page fifteen. Each page displays specific information and control commands for the mixer.

The default page on power-up of the unit is page zero (00). To "turn the page," enter the desired page number by pushing the **control button** next to the **control button label GET PAGE**. If a **template cell** has already been selected, push the blue **CANCEL** key on the **TouchPanel**. This action will deselect the **template cell** and display the default **GET PAGE control button label**.

After the **GET PAGE control button** is activated, the **VCP** will display an **ENTER PAGE # control button label**. Press the two-digit number of the desired page on the **TouchPanel's numeric keypad** (i.e., 01, 02, etc.), and the currently displayed page will be replaced with the page number entered.

SECTION 4-10

ENTER OR CHANGE NUMERIC VALUE

For most applications of the **PanelMate 3000 VCP**, timer values for the mixer run cycle must be entered. Enter numeric values by selecting the **template cell** for the value to be changed. When the **template cell** is activated, the screen will display the control button label **CHANGE VALUE**. Touch this control button to change the timer's numeric value. After the **CHANGE VALUE** control button is touched, the control button label **ENTER** will be displayed. Enter the new number using the **numeric keypad** on the **TouchPanel**. Be sure to enter decimal points as needed.

NOTE: All time values are entered in one hundredths of a minute--not in minutes and seconds. Enter 1 minute and 45 seconds as 1.75 minutes.

The value entered will be shown on the **control button label** next to the bottom control button. If this value is correct, touch the **ENTER** control button to enter the value into the system.

ADJUST SCREEN CONTRAST

NOTE: The following information is for grayscale VCPs only. The contrast on color screens cannot be changed.

The **PanelMate 3000 VCP** screen contrast can be changed at any time by using the **Adjust Screen Contrast** feature. To adjust the contrast, select the control button labeled **GET PAGE** from the default control buttons. When the next set of control buttons appears, select the control button labeled **MORE BUTTONS**. When the next set of control buttons appears, select the control button labeled **ADJUST CONTRAST**. Use the control buttons labeled **LIGHTER** and **DARKER** to adjust the display contrast.

To save the new setting, press the control button labeled **SAVE SETTING**. To revert to the previous contrast, press the **CANCEL** key. The **PanelMate 3000 VCP** will return to the last saved contrast level after power is cycled to the unit.

SECTION 4-10

ALLEN-BRADLEY PANELVIEW® SERIES 550 4-10-3

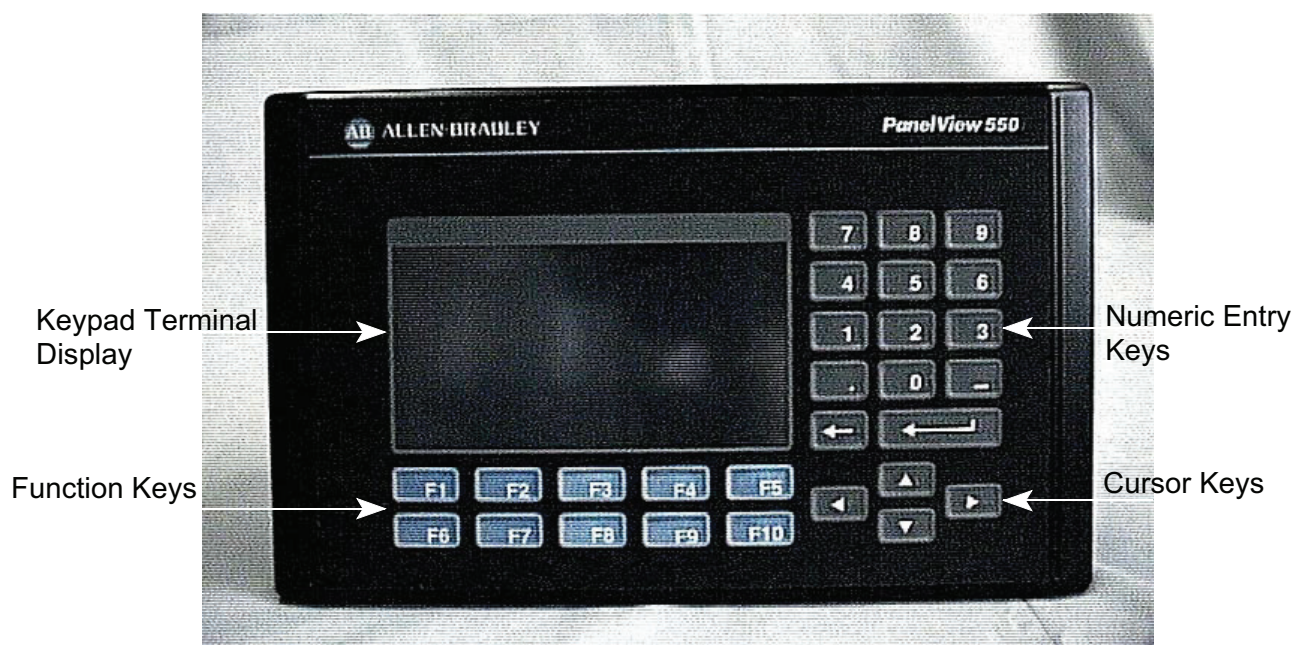
Your **Peerless** mixer can be equipped with an optional **Allen-Bradley PanelView® Series 550 VCP**. This section covers basic operation of the **Allen-Bradley PanelView 550**.

The **Allen-Bradley PanelView 550 VCP** is an operator-interface device for use with **PLCs**. The **PanelView 550 VCP** takes the place of conventional control panel components like indicator lights, pushbuttons, timers and digital displays such as thermometers. When mounted on the mixer, the **PanelView 550** is completely sealed against dust, dirt, oil and moisture. It has a NEMA 4X rating and will withstand moderate wash down with low-pressure water.

The front panel of the **PanelView 550** unit contains **function keys**, **cursor keys**, **numeric entry keys**, and **keypad terminal display**. See the following page for an explanation of these terms.

Figure 4-10-5 shows a typical first page displayed when the **Allen-Bradley PanelView 550** is powered up.

ALLEN-BRADLEY PANELVIEW® 550 VCP FIGURE 4-10-5



SECTION 4-10

When your mixer's power is turned on, the **Allen-Bradley PanelView 550** unit will perform various system diagnostics on itself to make sure it is working properly. If a problem occurs, turn the power off and on again. If the **VCP** reports the same problem, call your local **Allen-Bradley** distributor or the Engineering Department at **Peerless Machinery Corp.**

EXPLANATION OF PANELVIEW 550 CONTROLS AND DISPLAYS

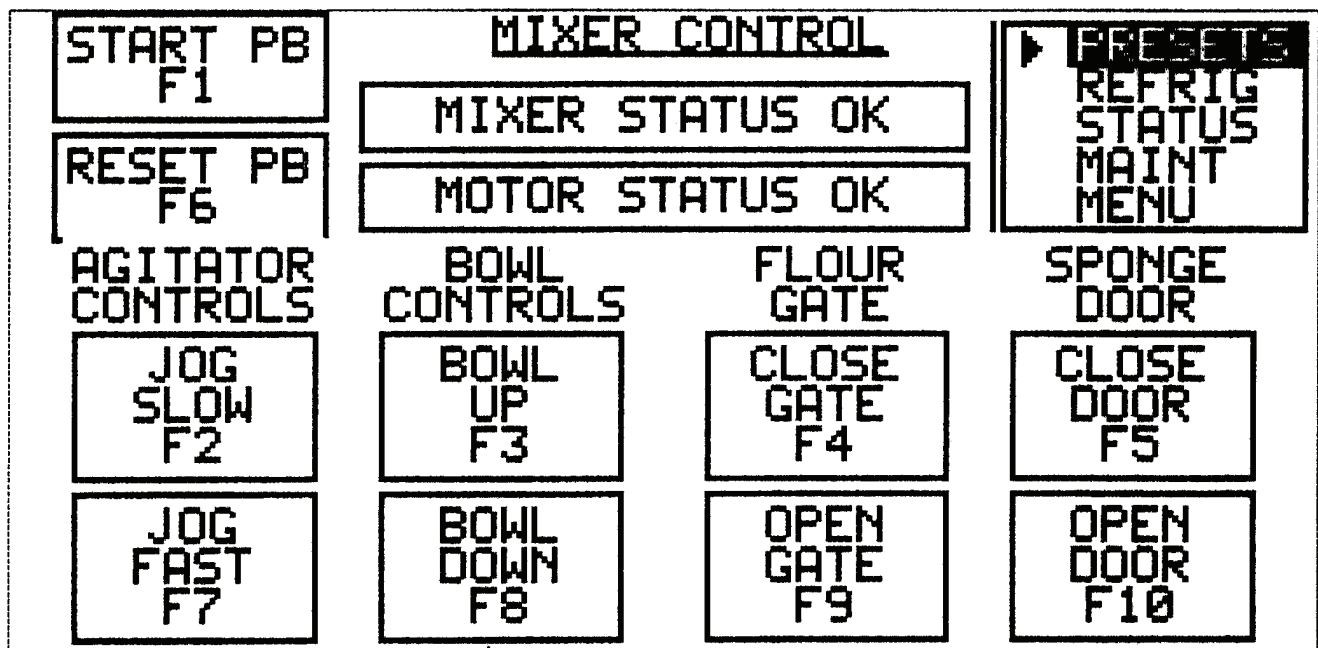
- | | | |
|---------------------------------|---|---|
| Function Keys. | ✓ | Use function keys on keypad terminals to initiate functions shown on terminal's display. |
| Cursor Keys. | ✓ | Use cursor keys to move cursor over displayed lists, to select a numeric entry item or to enter configuration mode. |
| Numeric Entry Keys. | ✓ | Use numeric entry keys [0] to [9] to enter numeric values. |
| | ✓ | Use decimal key [.] to enter a decimal point. |
| | ✓ | Use minus key [-] to enter a negative value. |
| | ✓ | Use backspace key [←] to clear previously entered digits. |
| | ✓ | Press enter key [↵] after a value has been correctly entered. |
| Keypad Terminal Display. | ✓ | On keypad terminals , initiate function of displayed items such as ON or OFF, by pressing one of the function keys [F1] through [F10] . |

SECTION 4-10

As soon as the mix cycle is started, the **STATUS SCREEN** will be displayed. When mixer operation is complete, the **VCP** will automatically display the **MIXER CONTROL SCREEN**. See Figure 4-10-6.

ALLEN-BRADLEY PANELVIEW® 550 VCP MIXER CONTROL SCREEN

FIGURE 4-10-6



SECTION 4-10

TO SELECT A TEMPLATE CELL

Press the **function key** that corresponds to the terminal display function you wish to select. See **Figure 4-10-5** or **4-10-6**.

For example, pressing the **[F10]** key in **Figure 4-10-5** will have no effect because no terminal display function has been assigned to it. Pressing the **[F10]** key in **Figure 4-10-6** will open the **flour gate**. Pressing the **[F10]** key in **Figure 4-10-7** will bring up the **MENU SCREEN** page.

ENTER OR CHANGE NUMERIC VALUE

For most applications of the **Allen-Bradley PanelView 550**, timer values for the mixer run cycle must be entered. See **Figure 4-10-7** for an example of the **PRESET TIMERS** page. Notice the **function key** assigned to each timer value. Press the **function key** corresponding to the timer you wish to change, and a numeric input screen will appear. After entering the new numbers, press the **enter key** to return to the **PRESET TIMERS** page, and the new value will appear in the **Preset Time** column. Pushing the **RESET TIMERS** button enters the new value into the controller.

NOTE:	All time values are entered in hundredths of a minute--not in minutes and seconds. For example, enter 1 minute and 45 seconds as 1.75 minutes.
--------------	--

SECTION 4-10

ALLEN-BRADLEY PANELVIEW® 550 VCP PRESET TIMERS PAGE FIGURE 4-10-7

RESET TIMERS F7	PRESET TIMERS			SCREEN SEL
	ACTUAL TIME	(MIN)	PRESET TIME	
SLOW MIX	##.##	F1	##.##	MIXER CONTROL F8
FAST MIX 1	##.##	F2	##.##	
FAST MIX 2	##.##	F3	##.##	STATUS SCREEN F9
REFRIG DELAY	##.##	F4	##.##	
REFRIG ON	##.##	F5	##.##	MENU SCREEN F10
DOUGH TEMP	###. #	F6	###. #	

SECTION 4-10

MULTIPLE PRODUCT MIXER CONTROL FEATURE 4-10-4

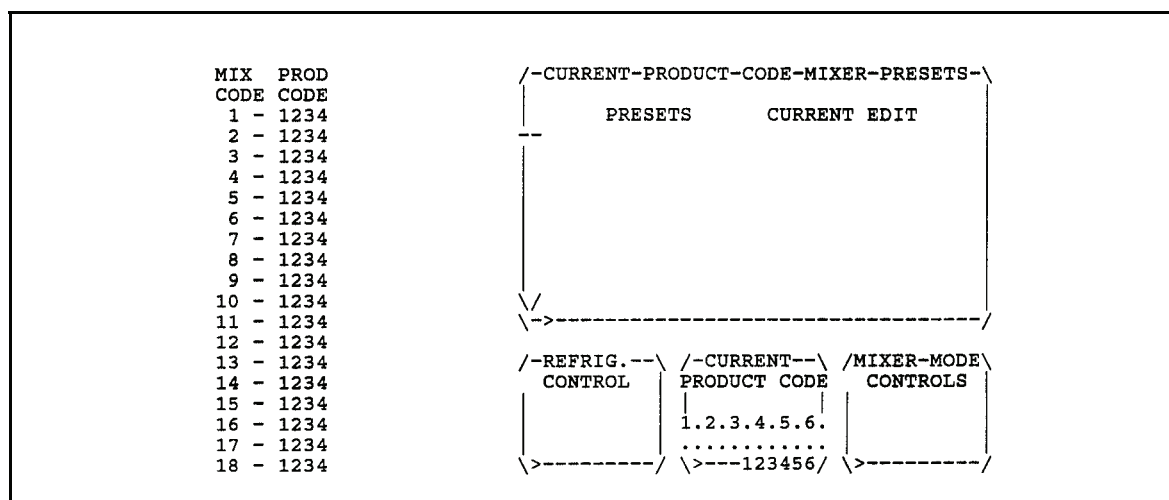
The **Multiple Product Mixer Control (MPMC) Feature**, available only on the **Cutler-Hammer IDT PanelMate 3000 VCP**, allows you to enter and edit timer preset values for multiple products.

See **Section 4-10-2** for instructions on how to use your **PanelMate 3000 VCP**.

The **MPMCF** uses two screens, or pages, of the **VCP** called **Current Mix Cycle Editor** (page 3) and **Master Product Code Editor** (page 4).

Use the **Current Mix Cycle Editor** page to run the mixer, call up product codes, and make changes to the current running cycle. Changes made to preset timer values on this page are not permanent and do not affect the **Master Mix Cycle Register**.

CURRENT MIX CYCLE EDITOR PAGE FIGURE 4-10-8



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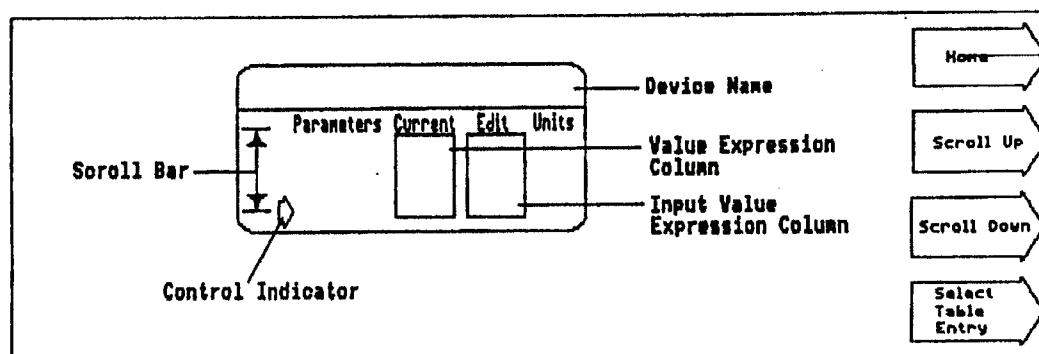
MASTER PRODUCT CODE EDITOR PAGE FIGURE 4-10-9

MIX CODE	PROD CODE	--MASTER-PRODUCT-CODE-MIXER-PRESETS--	
		PRESETS	CURRENT EDIT
1 -	1234		
2 -	1234		
3 -	1234		
4 -	1234		
5 -	1234		
6 -	1234		
7 -	1234		
8 -	1234		
9 -	1234		
10 -	1234		
11 -	1234		
12 -	1234		
13 -	1234		
14 -	1234		
15 -	1234		
16 -	1234		
17 -	1234		
18 -	1234		

/-MIX-CODE-ENTRY		/-PRODUCT CODE	
1.2.3.4.5.6.		1.2.3.4.5.6.	
.....			

Use the **MASTER PRODUCT CODE MIXER PRESET** table template to set up and edit the **Master Mix Cycle Register**. Enter or edit presets for each product on this page. Each entry preset has a current value, edit value and a units field. The template has built-in scroll capabilities to allow you to select the desired entry preset item. A scroll indicator built into the left side of the template indicates **TOP** and **BOTTOM** of the entire field. **Control buttons** on the right of the screen permit you to scroll up/down, return to home or open a selected field for numeric entry. **Figure 4-10-10** represents the **MASTER PRODUCT CODE MIXER PRESET** table template.

MASTER PRODUCT CODE MIXER PRESET TABLE TEMPLATE FIGURE 4-10-10



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Each preset can be reached with the scroll keys. The following table shows what each preset represents.

MASTER PRODUCT CODE PRESET TABLE 4-10-11

1st Preset:	Product Code	Numeric Entry	
2nd Preset:	Low Speed Timer	Numeric Entry	See Note 1 Below
3rd Preset:	1st Fast Speed Timer	Numeric Entry	See Note 1 Below
4th Preset:	2nd Fast Speed Timer	Numeric Entry	See Note 1 Below
5th Preset:	Refrig Delay Timer	Numeric Entry	See Note 1 Below
6th Preset:	Refrig On Timer	Numeric Entry	See Note 1 Below
7th Preset:	Dough Temp	Numeric Entry	See Note 2 Below

NOTE 1: Make all entries in hundredths (0.00) of minutes: Enter 1 minute and 30 seconds as 1.5 minutes. The **PanelMate 3000 VCP** accepts numeric entries between zero (0) and 60 minutes. An entry over 60 or less than zero (0) will not change current preset.

NOTE 2: Dough temperature setpoint. The **PanelMate 3000 VCP** accepts entries between 40° F (4.5 °C) and 120° F (49 °C). An entry over 120° F (49 °C) will not change current preset; an entry of less than 40° F (4.5 °C) will change current preset to 40° F (4.5 °C).

TO ENTER MASTER PRODUCT CODE PRESET VALUES

1. Go to the **Master Product Code Editor** page.
2. Choose one of the available **Mix Codes** and do the following:
 - a) Select the **MIX CODE ENTRY** template cell.
 - b) Push the **CHANGE VALUE** button.
 - b) Enter the numeric value corresponding to the **Mix Code** desired to create or edit.
 - d) See the selected **Mix Code** displayed in the **MIX CODE ENTRY** template. The **MASTER PRODUCT CODE MIXER PRESETS** template will also display selected **Mix Code** recipe values.

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3. Select the **template cell MASTER PRODUCT CODE MIXER PRESETS** and edit recipe values.
4. **Product Code:** Enter a four-digit **Product Code** from zero (0) to 9999.
5. **Low Speed Timer:** Set desired time on the **Low Speed Timer**.
6. **1st Fast Speed Timer:** Set desired time on the **1st Fast Speed Timer**.
7. **2nd Fast Speed Timer:** Set desired time on the **2nd Fast Speed Timer**.
8. **Refrig Delay Timer:** Set desired time on the **Refrigeration Delay Timer**.
9. **Refrig On Timer:** Set desired time on the **Refrigeration On Timer**.
10. **Dough Temp:** Set desired **Dough Temperature**.

NOTE: MPMCF option includes the temperature control option as described in **Section 4-5**.

11. To store an edited **Mix Code**, select the **template cell PRODUCT CODE** and push **UPDATE MASTER PRODUCT CODE FILE** control button.

TO USE AN ENTERED MASTER CODE MIX PRESET

1. Go to the **Current Mix Cycle Editor** page.
2. Select the **CURRENT PRODUCT CODE** template.
3. Push the **CHANGE VALUE** button.
4. Enter the **Mix Code** you wish to load.
5. New **Mix Code** preset values will be displayed on the screen.

NOTE 1: Current **Mix Code** preset values can be edited, but editing them on **Current Mix Cycle Editor** page will not change **Master Product Code** preset values.

NOTE 2: Pushing **PRODUCT CODE PRESET** button under the **Reset Controls** template will reset product code back to **MASTER PRODUCT CODE PRESET** values.

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VCP CONTROLS SUMMARY CHART 4-10-12

Following is a list of operator controls and displays specific to **Peerless** mixers equipped with **VCPs**.

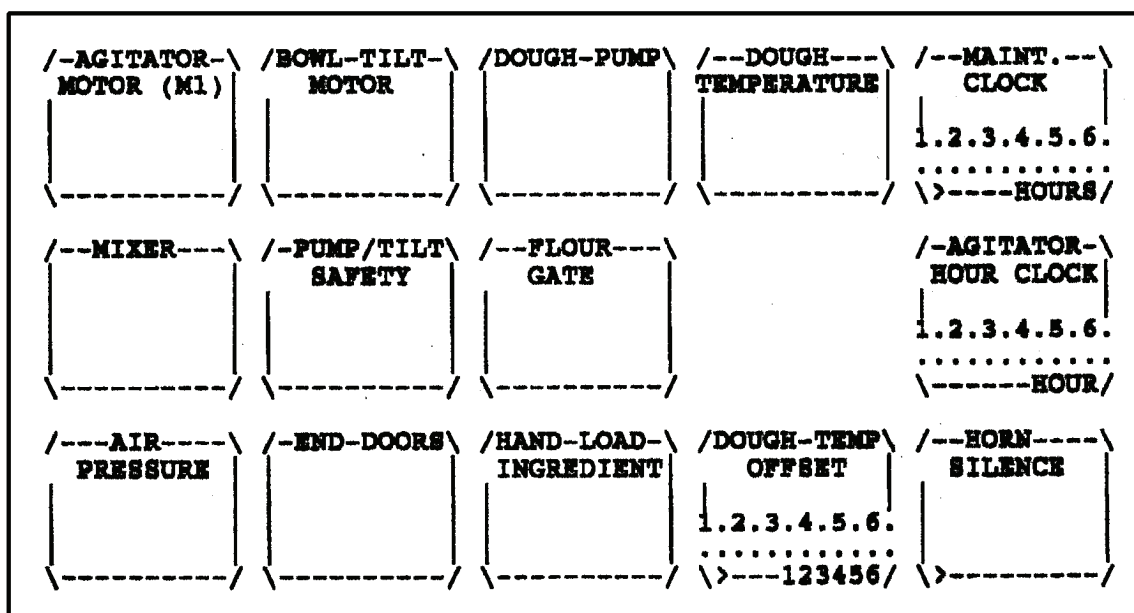
COMPONENT NAME	COMPONENT TYPE	FUNCTION
1ST FAST	Timer	Displays 1st Fast Speed time remaining. When 1ST Fast Speed timer times out to zero (0), mixer will stop.
2ND FAST	Timer	Displays 2nd Fast Speed time. 2ND FAST SPEED timer starts timing down when START button is pushed and no time remains on 1ST FAST SPEED timer. When 2ND FAST SPEED timer times out to zero (0), mixer will stop.
AGITATOR MOTOR (M1)	Display	Displays status of agitator motor: Low Speed On, Fast Speed On, Off .
END DOORS	Display	Displays status of right-hand end doors: Open, Closed .
FLOUR GATE STATUS	Display	Displays status of flour gate : Closed, Open, Jammed .
DOUGH TEMP OFFSET	Display	Value 1: displays dough temperature .
	Numeric Control	Value 2: Displays offset of dough temperature reading and main display.
AGITATOR HOURS	Clock	Displays total hours agitator motor has run.
SET MAINT CLOCK	Clock	Value 1: Displays hours remaining before general maintenance is required
	Numeric Entry	Value 2: Displays setpoint or set for maintenance clock counter. Numeric input values between zero (0) and 300.
REMAINING MAINT CLOCK HOURS	Clock	Displays hours remaining before warning display indicates maintenance clock has timed out. Pushing MAINTENANCE CLOCK RESET button with SAFETY button resets maintenance clock.

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VCP DIAGNOSTICS SCREEN 4-10-5

The **VCP's diagnostics screen** displays current mixer operating conditions and alerts the operator of any alarms or problems with the mixer. See **Figure 4-10-13** for a typical layout of the **diagnostics screen**. For more information about diagnostics screen functions, see a summary of the diagnostics screen functions following **Figure 4-10-13**.

DIAGNOSTICS SCREEN GENERAL PAGE LAYOUT FIGURE 4-10-13



(Page 4 on VCP)

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DIAGNOSTICS SCREEN FUNCTIONS SUMMARY CHART 4-10-14

COMPONENT NAME	COMPONENT TYPE	FUNCTION
AGITATOR MOTOR (M1)	(No Control)	Displays status of agitator motor: Low Speed On, High Speed On, Overload Tripped Alarm, Off.
BOWL TILT	(No Control)	Displays status of bowl and tilt motor: MOTOR Overload Tripped Alarm, Bowl Down, Bowl In Hand Load Position, Bowl Open, Bowl Closed.
DOUGH PUMP (OPTIONAL)	(No Control)	Displays status of dough pump: On, Off, Drive Fault, Motor Door Alarm
DOUGH TEMP	(No Control)	Displays status of dough temperature: Below Setpoint, Dough Hot Alarm, Above Setpoint.
MAINT CLOCK		Displays hours remaining before general maintenance is required.
	MAINT CLOCK RESET	Resets maintenance clock.
MIXER	(No Control)	Displays status of mixer: Maintenance Required Soon, Holding For Pump, In Hold Mode, Auto Mode, Manual Mode.
PUMP/TILT SAFETY (OPTIONAL)	(No Control)	Displays status of tilt safety gate: Gate Closed, Gate Open Alarm, Gate Open Stop Pump.
INGREDIENT DOOR (OPTIONAL)	(No Control)	Displays status of safety ingredient door: Closed, Open, Jammed Alarm.
FLOUR GATE	(No Control)	Displays status of flour gate: Closed, Open, Jammed Alarm.

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COMPONENT NAME	COMPONENT TYPE	FUNCTION
AGITATOR HOUR CLOCK	(No Control)	Displays total hours agitator motor has run.
AIR PRESSURE	(No Control)	Displays air pressure status: Low Alarm, Good.
END DOORS	(No Control)	Displays status of right-hand end doors: Closed, Open Alarm.
HAND LOAD (OPTIONAL)	(No Control)	Displays status of hand load ingredient: Delayed Warning Alarm, Added.
DOUGH TEMP	(No Control) Numeric Control	Value 1: Displays dough temperature Value 2: Displays offset of actual temperature probe reading and temperature displayed. Numeric values between -10 and 10 are accepted. Entered values greater than 10 or less than -10 will not change existing entry.
HORN SILENCE	HORN SILENCE	Silences warning horn when pushed.

SECTION 4-10**MAINTENANCE CLOCK AND AGITATOR HOURS TIMER 4-10-6**

The **VCP's AGITATOR HOURS** timer and **MAINTENANCE CLOCK** are internal **PLC** timers. **AGITATOR HOURS** logs accumulated agitator-run hours, and the **MAINTENANCE CLOCK** is used by maintenance personnel to help schedule general mixer maintenance.

The **AGITATOR HOURS** timer operates only when the agitator is running. The timer begins at zero (0) when the mixer is first started and accumulates the number of hours the agitator has run over the life of the mixer. *Accumulated agitator-run hours will be lost if the **PLC** software is written over.* See the note at the beginning of **Section 4-10**.

To set the **MAINTENANCE CLOCK**, go to page nine on the **VCP** screen. Push the **PRESET** button and enter the desired hours in timer values between zero (0) and 300. The timer will begin with the number entered and time down to zero (0). To start the **REMAINING MAINT CLOCK HOURS**, go to the diagnostics page and push the **RESET** button simultaneously with the **SAFETY** button.

The **REMAINING MAINT CLOCK HOURS** function operates only when the agitator is running. Once timer values have been entered, the timer displays hours remaining on the **MAINTENANCE CLOCK** before the timer times out to zero (0). When the timer times out, the **MAINTENANCE REQUIRED SOON** warning flashes on each page of the **VCP** screen until the **MAINTENANCE CLOCK** is reset. The warning message does not affect mixer operation.

NOTE: **Peerless** strongly recommends that scheduled maintenance be performed when **MAINTENANCE REQUIRED SOON** message flashes on **VCP** screen. *Do not reset timer before general maintenance has been performed on mixer.*